

STUDY-UNI · UCC MARKING SCHEME

MA1100

Introductory Mathematics for Business I

Full Model Answers & Mark Allocation

Examination Session: Summer 2025 · **Paper:** 1

Module Code: MA1100 · **Duration:** 1.5 hours

Instructions to Candidates: Answer Question 1 and two other questions.

Prepared by: Study-Uni

Question 1

Total: 30 marks

(a) [5 marks]

Solve for n in the equation $5(2/3)^n = 7/8$. Give your answer correct to one decimal place.

FULL MODEL ANSWER

Isolate the exponential term first:

$$5(2/3)^n = 7/8 \Rightarrow (2/3)^n = 7/40$$

Take **ln** of both sides and use the power rule $n \ln(2/3) = \ln(7/40)$:

$$n = \ln(7/40) / \ln(2/3) = (-1.7440) / (-0.4055)$$

$$n = 4.3 \text{ (to 1 decimal place)}$$

Checknote: A common error is taking logs before isolating the exponential term, which drags the coefficient 5 into the log incorrectly. Always isolate $(2/3)^n$ first.

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	5	Correctly isolates $(2/3)^n = 7/40$, AND takes logs of both sides correctly, AND arrives at $n = 4.3$.
Good	3	Correct method (isolate then take logs) with one arithmetic slip in the final value.
Partial	2	Correctly isolates $(2/3)^n = 7/40$ but does not complete the log solution, OR uses an incorrect log base without recovering.
Minimal	0	No valid isolation of the exponential term, or a purely guessed numeric answer with no working shown.

(b) [5 marks]

€25,000 is invested at a rate of 4.8% per annum compounded yearly. What is the value of the investment after 12 years?

FULL MODEL ANSWER

Apply the compound interest formula $A = P(1 + r)^t$ with $P = €25,000$, $r = 0.048$, $t = 12$:

$$A = 25\{, \}000 \times (1.048)^{12}$$

$$A = €43,880.89 \text{ (to the nearest cent)}$$

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	5	Correct formula with correct substitution AND correct final value €43,880.89 (± €0.01).
Good	3	Correct formula and substitution, minor rounding/calculator slip in the final figure.
Partial	2	Correct formula stated with correct values identified (P, r, t) but not correctly evaluated.
Minimal	0	Incorrect formula (e.g. simple interest) or no coherent attempt.

(c)(i) [2 marks]

Calculate the Annual Percentage Rate (APR) for an interest rate of 7.2% compounded monthly, to 4 decimal places.

FULL MODEL ANSWER

$APR = (1 + r/m)^m - 1$, with $r = 0.072$, $m = 12$:

$$APR = (1 + 0.072/12)^{12} - 1 = (1.006)^{12} - 1$$

$$APR = 7.4424\%$$

Mark Allocation — (c)(i)

Tier	Marks	Criteria
Full credit	2	Correct formula, correct substitution, AND APR = 7.4424% to 4dp.
Partial	1	Correct formula and substitution with a rounding or calculator slip in the final percentage.
Minimal	0	Incorrect or missing formula.

(c)(ii) [2 marks]

Calculate the Annual Percentage Rate (APR) for an interest rate of 7.2% compounded continuously, to 4 decimal places.

FULL MODEL ANSWER

For continuous compounding, $APR = e^r - 1$, with $r = 0.072$:

$$APR = e^{0.072} - 1$$

$$APR = 7.4655\%$$

Checknote: 7.2% continuously compounded gives a very slightly higher APR than monthly compounding — this ordering (continuous $\geq$ monthly $\geq$ nominal) is a useful self-check across (c)(i) and (c)(ii).

Mark Allocation — (c)(ii)

Tier	Marks	Criteria
Full credit	2	Correct continuous-compounding formula AND APR = 7.4655% to 4dp.
Partial	1	Correct formula with a rounding or calculator slip.
Minimal	0	Uses the discrete-compounding formula instead of e^r , or no coherent attempt.

(d)(i) [2 marks]

Find the derivative of $f(x) = 3x^2 + \sqrt{x}$ and write your answer in its simplest form.

FULL MODEL ANSWER

Differentiate term by term, writing $\sqrt{x}$ as $x^{1/2}$:

$$f(x) = 3x^2 + x^{1/2}$$

$$f'(x) = 6x + 1/(2\sqrt{x})$$

Mark Allocation — (d)(i)

Tier	Marks	Criteria
Full credit	2	Correct power-rule differentiation of both terms AND simplified final answer $6x + 1/(2\sqrt{x})$.
Partial	1	Correct differentiation of one term only, or correct derivative left unsimplified.
Minimal	0	Incorrect application of the power rule.

(d)(ii) [2 marks]

Find the derivative of $f(x) = e^x(x - 1)$ and write your answer in its simplest form.

FULL MODEL ANSWER

Apply the **product rule**, $u = e^x$, $v = (x-1)$, so $u' = e^x$, $v' = 1$:

$$f'(x) = e^x(x-1) + e^x(1) = e^x[(x-1) + 1]$$

$$f'(x) = xe^x$$

Mark Allocation — (d)(ii)

Tier	Marks	Criteria
Full credit	2	Correct product rule setup AND simplifies fully to xe^x .
Partial	1	Correct product rule setup but does not simplify to xe^x .
Minimal	0	Product rule not applied, or incorrect derivative of e^x .

(d)(iii) [2 marks]

Find the derivative of $f(x) = 2^x/3^x$ and write your answer in its simplest form. (Note: if $y = a^x$ then $dy/dx = a^x \ln a$.)

FULL MODEL ANSWER

Rewrite as a single exponential base first, since $2^x/3^x = (2/3)^x$:

$$f(x) = (2/3)^x$$

Apply the given rule $dy/dx = a^x \ln a$ with $a = 2/3$:

$$f'(x) = (2/3)^x \ln(2/3)$$

Checknote: $\ln(2/3)$ is negative, so this derivative is negative for all x — a quick sanity check, since $(2/3)^x$ is a decreasing function.

Mark Allocation — (d)(iii)

Tier	Marks	Criteria
Full credit	2	Correctly rewrites as $(2/3)^x$ AND applies $a^x \ln a$ correctly.
Partial	1	Applies the rule to 2^x and 3^x separately via the quotient rule and reaches an equivalent but unsimplified correct expression.
Minimal	0	Does not apply the given differentiation rule correctly.

(d)(iv) [2 marks]

Find the derivative of $f(x) = \sqrt[3]{(x+1)}$ and write your answer in its simplest form.

FULL MODEL ANSWER

Write as a power and apply the **chain rule**:

$$f(x) = (x+1)^{1/3}$$

$$f'(x) = (1/3)(x+1)^{-2/3} \times 1$$

$$f'(x) = 1 / [3(x+1)^{2/3}]$$

Mark Allocation — (d)(iv)

Tier	Marks	Criteria
Full credit	2	Correct chain rule application AND simplified final form.
Partial	1	Correct power/chain rule set-up with an index-law slip in simplification.
Minimal	0	Chain rule not applied (inner derivative omitted).

(e) [8 marks]

Using implicit differentiation, find dy/dx for $4x^2 - y^2 + 3xy = 21$. Given that $A = (2,1)$ is a point on the graph, find the slope of the tangent line at A. Hence, find the equation of the tangent line at A.

FULL MODEL ANSWER

Differentiate every term with respect to x , treating y as a function of x . The term $3xy$ needs the **product rule**, and y^2 needs the **chain rule**:

$$8x - 2y(dy/dx) + 3y + 3x(dy/dx) = 0$$

Collect the dy/dx terms on one side:

$$(dy/dx)(3x - 2y) = -8x - 3y$$

$$dy/dx = (8x + 3y) / (2y - 3x)$$

Slope at A(2,1): substitute $x = 2$, $y = 1$:

$$dy/dx = (8(2) + 3(1)) / (2(1) - 3(2)) = 19 / (-4)$$

$$\text{slope} = -19/4$$

Tangent line at A(2,1): use point-slope form $y - y_1 = m(x - x_1)$:

$$y - 1 = -19/4 (x - 2)$$

$$y = -19/4 x + 10.5$$

Checknote: The most common slip is differentiating $3xy$ as just $3y$ (forgetting the product rule) or y^2 as just $2y$ (forgetting the chain rule dy/dx factor). Both must be present for full credit.

Mark Allocation — (e)

Tier	Marks	Criteria
Full credit	8	Correct implicit differentiation of all three terms (product rule on $3xy$, chain rule on y^2) AND correct general dy/dx AND correct slope at A ($-19/4$) AND correct tangent line equation.
Good	6	Correct dy/dx and correct slope at A, but an error in forming or simplifying the final tangent line equation.
Partial	4	Correct differentiation of each term individually but an error collecting dy/dx terms, so the general dy/dx expression is wrong — slope and tangent line follow through correctly from the student's own (incorrect) expression.
Minimal	1	Product rule or chain rule omitted on one or more terms (e.g. treating y^2 or $3xy$ as if y were a constant).

Question 1 — Total

30/30 marks confirmed

Question 2

Total: 25 marks

(a) [5 marks]

The supply function is $P_s = (1/6)Q + 1$ and the demand function is $P_d = 7 - (1/3)Q$. If P is on the vertical axis and Q is on the horizontal axis, write down the slope of the supply function and the slope of the demand function.

FULL MODEL ANSWER

Both functions are already written as $P = (\text{slope}) \cdot Q + \text{intercept}$, so the slopes read off directly:

Function	Equation	Slope (dP/dQ)
Supply	$P_s = (1/6)Q + 1$	1/6
Demand	$P_d = 7 - (1/3)Q$	-1/3

Checknote: Because P (not Q) is on the vertical axis, the slope requested is dP/dQ , which is simply the coefficient already in front of Q in each equation — no rearrangement needed.

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	5	Both slopes correctly identified: supply = 1/6, demand = -1/3, with correct sign on the demand slope.
Partial	3	One slope correct, or both magnitudes correct but the negative sign on demand omitted.
Minimal	0	Slopes not correctly extracted from the given equations.

(b) [6 marks]

Determine the equilibrium price and quantity.

FULL MODEL ANSWER

At equilibrium, $P_s = P_d$:

$$(1/6)Q + 1 = 7 - (1/3)Q$$

$$(1/2)Q = 6 \Rightarrow Q = 12$$

Substitute back into either equation to find P:

$$P = (1/6)(12) + 1 = 2 + 1 = 3$$

EQUILIBRIUM

$$Q^* = 12, P^* = €3$$

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	6	Correctly sets $P_s = P_d$, solves for $Q = 12$ AND $P = €3$, with both values stated clearly.
Good	4	Correct method and correct Q, with an arithmetic slip carried into the final P value (or vice versa).
Partial	2	Correct equation set up ($P_s = P_d$) but not solved through to numeric values.
Minimal	0	Incorrect equilibrium condition used.

(c) [8 marks]

If a subsidy of €2.00 is given on the sale of each item, what will be the new equilibrium price and quantity be?

FULL MODEL ANSWER

A per-unit subsidy paid to sellers lowers the price at which they are willing to supply any given quantity — it shifts the supply curve **down** by the amount of the subsidy:

$$P_{s(\text{new})} = (1/6)Q + 1 - 2 = (1/6)Q - 1$$

Set the new supply curve equal to the (unchanged) demand curve:

$$(1/6)Q - 1 = 7 - (1/3)Q$$

$$(1/2)Q = 8 \Rightarrow Q = 16$$

Substitute into the demand curve to get the price consumers pay:

$$P = 7 - (1/3)(16) = 7 - 16/3 = 5/3$$

New $Q^* = 16$, new price paid by consumers = €1.67 (to the nearest cent)

The price producers actually *receive* per unit is this consumer price plus the €2 subsidy: €1.67 + €2.00 = **€3.67** — this figure is needed for part (d).

Checknote: A subsidy is the mirror image of a tax: a tax shifts supply up (sellers need more to break even), a subsidy shifts supply down (sellers are compensated, so they'll supply the same Q for a lower market price).

Mark Allocation — (c)

Tier	Marks	Criteria
Full credit	8	Correctly shifts the supply curve down by €2 AND solves the new equilibrium AND correctly reports $Q = 16$, consumer price $\approx$ €1.67 (with producer's received price identified as €3.67).
Good	6	Correct direction of shift and correct new equilibrium Q and consumer price, but the distinction between consumer price and producer's received price not drawn out.
Partial	3	Correctly identifies that supply shifts by €2 but shifts it the wrong direction (up instead of down), carrying through consistently.
Minimal	0	Subsidy applied to the demand curve instead of supply, or no shift applied at all.

(d) [6 marks]

In what ratio is the subsidy shared between the producer and the consumer?

FULL MODEL ANSWER

Compare each side's price before and after the subsidy (using the original equilibrium price €3 from part (b) as the baseline):

Party	Price before	Price after	Benefit
Consumer pays	€3.00	€1.67	€1.33
Producer receives	€3.00	€3.67	€0.67
Total (must equal the €2.00 subsidy)			€2.00

Expressing the benefits as a ratio, producer : consumer:

$$0.67 : 1.33 \equiv 1 : 2$$

The consumer captures twice the benefit of the producer. This follows directly from the relative slopes in part (a): demand is steeper (less elastic, $|\text{slope}| = 1/3$) than supply (more elastic, $|\text{slope}| = 1/6$), and the flatter (more elastic) side of the market always captures the larger share of a subsidy.

Mark Allocation — (d)

Tier	Marks	Criteria
Full credit	6	Correctly computes both the consumer price fall (€1.33) and producer price rise (€0.67), AND confirms they sum to €2.00, AND states the ratio 1:2 (producer:consumer).
Good	4	Correct ratio 1:2 obtained, without explicit reference to the sum-to-€2 check or the elasticity explanation.
Partial	2	One of the two benefit amounts (consumer fall or producer rise) correctly computed, but the ratio not correctly formed.
Minimal	0	No coherent attempt to split the €2 subsidy between the two parties.

Question 2 — Total

25/25 marks confirmed

Question 3

Total: 25 marks

(a) [8 marks]

A person saves €600 a month for 40 years towards a pension, at an interest rate of 3.6% per annum compounded monthly. What will the value of the pension fund be in 40 years' time?

FULL MODEL ANSWER

Use the future value annuity formula from the formula sheet, $V_n = A_0[(1+i)^n - 1] / i$, with the monthly deposit $A_0 = €600$, monthly rate $i = 0.036/12 = 0.003$, and $n = 40 \times 12 = 480$ months:

$$V_{480} = 600 \times [(1.003)^{480} - 1] / 0.003$$

$$V_{480} = €642,321.42 \text{ (to the nearest cent)}$$

Checknote: Always convert BOTH the rate and the number of periods to the same frequency as the payments — monthly payments require a monthly i and a monthly n , never a mix of annual and monthly.

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	8	Correctly identifies $A_0 = 600$, $i = 0.003$, $n = 480$ AND applies the future value formula correctly AND reaches €642,321.42.
Good	6	Correct formula and correct i , n , but a slip in the final evaluation.
Partial	4	Correct formula identified but incorrect conversion of the annual rate/term to monthly (e.g. using $n = 40$ instead of 480, or $i = 0.036$ instead of 0.003).
Minimal	1	Wrong formula used (e.g. present value formula, or simple compound interest with no annuity structure).

(b) [9 marks]

The monthly repayments on a mortgage of €550,000 are €2,501. If the interest rate is 3.6% per annum compounded monthly, how many years does it take to pay off the mortgage?

FULL MODEL ANSWER

Use the present value annuity formula $V_0 = A_0[1 - (1+i)^{-n}] / i$, with $V_0 = €550,000$, $A_0 = €2,501$, $i = 0.036/12 = 0.003$. Solve for n :

$$550,000 = 2,501 \times [1 - (1.003)^{-n}] / 0.003$$

$$1 - (1.003)^{-n} = 550,000 \times 0.003 / 2,501 = 0.65974$$

$$(1.003)^{-n} = 0.34026$$

Take logs of both sides and solve for n :

$$-n \cdot \ln(1.003) = \ln(0.34026) \Rightarrow n = 359.88 \text{ months}$$

$$n \approx 360 \text{ months} = 30 \text{ years}$$

Checknote: n comes out to 359.88 rather than exactly 360 — this is expected, since the quoted monthly repayment of €2,501 is itself rounded. Rounding to 30 years is the correct, expected answer.

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	9	Correctly sets up the present value annuity formula with the given values AND correctly isolates $(1.003)^{-n}$ AND solves via logs to $n \approx 30$ years.
Good	7	Correct set-up and correct isolation of the exponential term, with a slip in the final log-solving step.
Partial	5	Correct formula identified and correctly substituted, but not successfully rearranged to isolate the exponential term.
Minimal	2	Uses the future value formula instead of present value, or does not attempt to solve for n via logarithms.

(c) [8 marks]

The repayments on a loan of €40,000 are €416 each month. Interest of 7.2% per annum is charged on the loan, compounded monthly. After 6 years, it is decided to pay the balance of the loan in one lump sum. What is the remaining balance on the loan after 6 years?

FULL MODEL ANSWER

The outstanding balance after k payments equals the loan's compounded original value minus the accumulated value of the payments already made:

$$B_k = P_0(1+i)^k - A_0[(1+i)^k - 1] / i$$

With $P_0 = €40,000$, $A_0 = €416$, $i = 0.072/12 = 0.006$, and $k = 6 \times 12 = 72$ payments made:

$$B_{72} = 40,000(1.006)^{72} - 416 \times [(1.006)^{72} - 1] / 0.006$$

$$B_{72} = €24,208.46 \text{ (to the nearest cent)}$$

Checknote: This is the same annuity mechanism as parts (a) and (b), applied to find a mid-term balance rather than a full future or present value — worth cross-referencing if either of those went wrong.

Mark Allocation — (c)

Tier	Marks	Criteria
Full credit	8	Correctly states the outstanding-balance formula (compounded original loan minus accumulated payments made) AND correctly substitutes $i = 0.006$, $k = 72$ AND reaches €24,208.46.
Good	6	Correct formula and correct substitution with a rounding or calculator slip in the final value.
Partial	4	Correctly computes the compounded value of the original €40,000 loan alone ($40,000 \times 1.006^{72}$), but does not subtract the accumulated value of payments made.
Minimal	1	Uses simple subtraction of nominal payments ($40,000 - 72 \times 416$) rather than a time-value-of-money approach.

Question 3 — Total

25/25 marks confirmed

Question 4

Total: 25 marks

4(a)(i) [5 marks]

A supply function is given by $Q = 80 - 2P - 0.5P^2$. Calculate the elasticity equation as a function of P .

FULL MODEL ANSWER

Price elasticity of supply is $E = (dQ/dP) \times (P/Q)$. First differentiate Q with respect to P :

$$dQ/dP = -2 - P$$

Then form the elasticity expression using the original $Q(P)$:

$$E(P) = (-2 - P) \times P / (80 - 2P - 0.5P^2)$$

Mark Allocation — 4(a)(i)

Tier	Marks	Criteria
Full credit	5	Correctly differentiates Q to get $dQ/dP = -2 - P$ AND correctly forms $E(P) = (dQ/dP)(P/Q)$ with the full expression for Q in the denominator.
Partial	3	Correct dQ/dP but elasticity formula assembled incorrectly (e.g. Q/P used instead of P/Q).
Minimal	0	Incorrect differentiation of Q with respect to P .

4(a)(ii) [5 marks]

Find the price elasticity of supply when the current price is $P = 30$.

FULL MODEL ANSWER

Substitute $P = 30$ into both dQ/dP and Q :

$$dQ/dP = -2 - 30 = -32$$

$$Q = 80 - 2(30) - 0.5(30)^2 = 80 - 60 - 450 = -430$$

$$E(30) = (-32) \times (30 / -430)$$

$$E(30) = 96/43 \approx 2.23$$

Checknote: $Q(30)$ is negative for this particular function — this is a feature of the algebra given in the question, not a sign the working has gone wrong. Follow the numbers through exactly as they fall.

Mark Allocation — 4(a)(ii)

Tier	Marks	Criteria
Full credit	5	Correctly substitutes $P = 30$ into both dQ/dP and $Q(P)$ AND reaches $E(30) \approx 2.23$, carried through consistently from the student's own (a)(i) expression.
Good	3	Correct substitution into dQ/dP and Q with an arithmetic slip in the final ratio.
Partial	1	Only one of dQ/dP or Q correctly evaluated at $P = 30$.
Minimal	0	No substitution attempted.

4(a)(iii) [2 marks]

What is the percentage change in supply if the price rises by 4%?

FULL MODEL ANSWER

By definition, $E \approx (\% \Delta Q) / (\% \Delta P)$, so $\% \Delta Q = E \times \% \Delta P$:

$$\% \Delta Q = 2.23 \times 4\%$$

$$\% \Delta Q \approx 8.93\%$$

Mark Allocation — 4(a)(iii)

Tier	Marks	Criteria
Full credit	2	Correctly applies $\% \Delta Q = E \times \% \Delta P$ using the student's own $E(30)$, reaching $\approx 8.93\%$.
Minimal	0	Formula not applied, or elasticity from (a)(ii) not correctly carried forward.

4(b)(i) [3 marks]

A manufacturer has fixed costs of €200 each week, and variable costs per unit $VC = 2Q - 36$, where Q is the quantity produced. Write down an expression in Q for the total cost function TC .

FULL MODEL ANSWER

Total cost is fixed costs plus total variable cost. Total variable cost is Q multiplied by the *per-unit* variable cost given:

$$TC = FC + Q \times VC = 200 + Q(2Q - 36)$$

$$TC = 2Q^2 - 36Q + 200$$

Checknote: $VC = 2Q - 36$ is given as a *per-unit* cost, so it must be multiplied by Q to get total variable cost — a frequent trap is adding it directly to fixed costs without that multiplication.

Mark Allocation — 4(b)(i)

Tier	Marks	Criteria
Full credit	3	Correctly multiplies Q by the per-unit VC expression AND adds fixed costs AND expands to $2Q^2 - 36Q + 200$.
Partial	1	Correct structure ($FC + Q \cdot VC$) but not expanded, or an expansion slip.
Minimal	0	Treats $VC = 2Q - 36$ as already the total variable cost (i.e. does not multiply by Q).

4(b)(ii) [3 marks]

Write down an expression in Q for the average cost function AC .

FULL MODEL ANSWER

Average cost is total cost divided by quantity:

$$AC = TC/Q = (2Q^2 - 36Q + 200) / Q$$

$$AC = 2Q - 36 + 200/Q$$

Mark Allocation — 4(b)(ii)

Tier	Marks	Criteria
Full credit	3	Correctly divides the TC expression from (b)(i) by Q term-by-term, reaching $2Q - 36 + 200/Q$.
Partial	1	Correct division method, carried through from an incorrect TC in (b)(i).
Minimal	0	AC not correctly formed as TC/Q .

4(b)(iii) [7 marks]

Find the value of Q that minimises the AC, and verify that it is a minimum.

FULL MODEL ANSWER

Differentiate AC with respect to Q and set equal to zero:

$$dAC/dQ = 2 - 200/Q^2$$

$$2 - 200/Q^2 = 0 \Rightarrow Q^2 = 100 \Rightarrow Q = 10 \text{ (taking the positive root, since quantity cannot be negative)}$$

Verify it is a minimum using the second derivative:

$$d^2AC/dQ^2 = 400/Q^3$$

$$\text{At } Q = 10: d^2AC/dQ^2 = 400/1000 = 0.4 > 0$$

CONFIRMED MINIMUM

Since the second derivative is positive at $Q = 10$, this is a genuine minimum ($AC = 2(10) - 36 + 200/10 = €4$ at this point).

Checknote: $Q = -10$ is mathematically also a solution of $Q^2 = 100$ but is economically meaningless (negative output), so it must be explicitly rejected, not just silently dropped.

Mark Allocation — 4(b)(iii)

Tier	Marks	Criteria
Full credit	7	Correctly sets $dAC/dQ = 0$ AND solves $Q = 10$ (rejecting the negative root) AND correctly computes $d^2AC/dQ^2 > 0$ at $Q = 10$ to confirm a minimum.
Good	5	Correct first-order condition and $Q = 10$, but the second-derivative (or equivalent) verification step is missing.
Partial	3	Correct differentiation of AC but an error solving $Q^2 = 100$ (e.g. keeping both roots without rejecting the negative one).
Minimal	1	AC not correctly differentiated.

Question 4 — Total

25/25 marks confirmed